THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** the container and the flame also differ along with the volume	Volume of water is a variable.
2	**(d)** a grass	Grasses show these features.
3	**(a)** starch and fat	Both tests are positive.
4	**(d)** P and R are magnets, Q is magnetic material	Repulsion is seen only between magnets.
5	**(b)** 160 cm	320 - 160.
6	**(a)** P	The pin falls first.
7	**(b)** 50 °C	Linear scale with equal spacing gives 50 °C.
8	**(b)** heat is used to change the state	Latent heat idea.
9	**(a)** magnet, add water and filter, evaporate filtrate	Each step removes one component.
10	**(d)** it does not grow or reproduce on its own	Many characteristics are absent.
11	**(d)** most oxygen was used	Burning consumed oxygen.
12	**(d)** 8 light-minutes	Approx.
13	**(a)** both depend on hot weather	A third factor links them.
14	**(b)** helps judge distance for hunting	Forward eyes give depth perception.
15	**(d)** they contain much sugar and few nutrients	Empty calories.
16	**(a)** iron is magnetic and copper is not	Only iron is attracted.
17	**(a)** time period is 1.8 s in both	T = 36/20 = 1.8 and 18/10 = 1.8.
18	**(b)** B	Translucent materials.
19	**(d)** 40 s	Twice slower.
20	**(b)** 100 g	Mass is conserved.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Two factors differ: light and water (1); We cannot tell which factor caused the difference (1).	1+1
22	Green stem carries out photosynthesis (1); Spines reduce water loss (1).	1+1
23	It contains starch (1); It also contains fat (1).	1+1
24	The steel table disturbs the needle (1); Move it to a wooden or non-magnetic surface (1).	1+1
25	3.6 km = 360000 cm (1); 360000 / 60 = 6000 steps (1).	1+1
26	Empty bottle has low average density (1); Sand makes it heavier (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Use same size, same wetness (1); Same place, time and temperature (1); Measure drying time for each (1).	1+1+1
28	Tree (1); Tap root goes deep (1); Net veins in leaves (1).	1+1+1
29	Lacks protein and vitamins (1); High in refined carbohydrates and fat (1); Add milk, dal, vegetables and fruit (1).	1+1+1
30	Roll bar in filings (1); Maximum cling near the ends (1); Little in the middle (1).	1+1+1
31	1.1 km = 1100 m (1); 1100 / 2.2 (1); 500 turns (1).	1+1+1
	OR: It is a metal (1); Shiny (lustrous) and sinks (1); Conducts heat along its length (1).	1+1+1
32	Atmospheric pressure is lower on hills (1); Water boils at a lower temperature (1); Therefore reading differs (1).	1+1+1
	OR: 4 x 30 = 120 L (1); 120/2000 (1); 6 per cent (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Hypothesis: dark cups cool faster (1); Same volume and temperature of water (1); Cups of different colour, same size and material (1); Measure temperature every 5 minutes (1); Plot results and conclude (1).	1+1+1+1+1
34	Aquatic: fish, whale (1); Terrestrial: cow, lion (1); Aerial: crow, eagle (1); Herbivore/carnivore/omnivore assigned (1); Features such as fins, legs, wings explained (1).	1+1+1+1+1
35	A: vitamin C, citrus fruits (1); B: vitamin A, carrot (1); C: iodine, iodised salt (1); D: vitamin D, sunlight and milk (1); E: iron, spinach (1).	1+1+1+1+1
36	Same type of clips (1); Same method to approach (1); Count clips lifted by each pole (1); Repeat three times (1); Compare averages (1).	1+1+1+1+1
	OR: Tree trunk: measuring tape or thread around (1); Then measure thread (1); Coin: stack 20 coins and measure (1); Divide by 20 (1); Pencil: ruler with zero at one end (1).	1+1+1+1+1
37	Same heat source and time (1); Measure temperature of the free end (1); Lowest rise is best (1); Use tongs and avoid burns (1); Rank and justify (1).	1+1+1+1+1
	OR: Same volume, same starting temperature (1); Same containers (1); Record temperature every 2 min (1); Plot a graph or table (1); Compare cooling (1).	1+1+1+1+1
38	Identical wet cloths (1); Place in three locations (1); Weigh at intervals (1); Greatest loss shows the fastest drying (1); Conclusion with reasons (1).	1+1+1+1+1
	OR: Keep both still (1); Mud settles, salt water stays clear (1); Filter both (1); Mud on paper, salt in filtrate (1); Evaporate filtrate for salt (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 30 g	Same as initial.
2	**(a)** pebble	It has one cell.
3	**(d)** adding compost	Compost adds humus.
4	**(b)** 60 kg	Mass does not change.
5	**(d)** compare the methods and repeat carefully	Differences in method may explain.
6	**(a)** many plants will fail to pollinate	Pollination would fall.
7	**(b)** fat	Protein test.
8	**(a)** always in the same direction with the same pole of a magnet	Opposite pole forms at the lifting end.
9	**(d)** 10.5 cm	12.5 - 2.
10	**(d)** its shape displaces much water	Hollow shape lowers average density.
11	**(a)** aluminium and plastic	Pan conducts, handle insulates.
12	**(c)** it changed to vapour and mixed with air	Matter is conserved.
13	**(d)** it makes tiny particles clump and settle	Coagulation.
14	**(d)** a living thing can stay inactive until conditions suit it	Life processes were only slowed down.
15	**(b)** more public transport	Fewer vehicles.
16	**(b)** Earth rotates on its axis	Tilt.
17	**(a)** more than one variable was changed	Only one variable should vary.
18	**(c)** frog	Frogs show this.
19	**(a)** Q contains starch	Iodine tests starch only.
20	**(d)** south-seeking pole	The end that points south is the south-seeking pole.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Temperatures are equal (1); The larger quantity contains more heat energy (1).	1+1
22	Vapour cannot escape (1); Air in the bag becomes saturated (1).	1+1
23	Some salt stuck to the filter or vessel (1); Handling losses (1).	1+1
24	It does not grow or reproduce (1); It does not respire or respond (1).	1+1
25	Oxygen about 21 per cent is used (1); Water enters to fill that space (1).	1+1
26	228/150 (1); about 1.5 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Magnet for iron (1); Water: wooden chips float, skim (1); Filter sand and chalk; evaporate salt (1).	1+1+1
28	No chlorophyll (1); Do not make their own food (1); Absorb food from dead matter (1).	1+1+1
29	21 per cent of 50 (1); 0.21 x 50 (1); 10.5 m^3 (1).	1+1+1
30	5000/10 (1); 500 days (1); Year/day (1).	1+1+1
31	Curiosity: why do things fall (1); Observation: fall of apple and Moon's motion (1); Reasoning: same force acts on both (1).	1+1+1
	OR: Colour can vary within a group (1); Unrelated animals may have the same colour (1); Features like body covering and habitat are better (1).	1+1+1
32	Calcium for bones (1); Proteins for growth (1); Vitamins like A and D (1).	1+1+1

	OR: Permanent magnet is present (1); A coil moves in its field (1); Produces sound (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Set A: dry seeds (1); Set B: seeds in water without air (boiled water with oil) (1); Set C: seeds in moist cotton at room temperature (1); Set D: moist seeds in the fridge (1); Only C germinates well (1).	1+1+1+1+1
34	List water use (1); List electricity use (1); Record fuel (1); Identify wastage (1); Suggest five actions (1).	1+1+1+1+1
35	Choose a scale for sizes (1); Place the Sun at the centre (1); Planets in order (1); Label features (1); Show orbits (1).	1+1+1+1+1
36	Two sets of same plants (1); One with music, one without (control) (1); Same water, soil, light (1); Measure height weekly (1); Compare and conclude (1).	1+1+1+1+1
	OR: Animals lose food and shelter (1); Plant cover declines (1); Plant more trees (1); Create national parks (1); Avoid pollution and hunting (1).	1+1+1+1+1
37	Cereal for energy (1); Dal for proteins (1); Vegetables for vitamins (1); Curd for calcium (1); Salad and pickle for fibre and flavour (1).	1+1+1+1+1
	OR: Roll magnet in filings (1); Filings gather at the ends (1); Fewer in middle (1); Count or compare quantity (1); Poles are strongest (1).	1+1+1+1+1
38	One reading may have error (1); Repeat several times (1); Take average (1); Avoid parallax (1); Use a good scale (1).	1+1+1+1+1
	OR: Take a steel foil (1); Place flat: sinks (1); Shape as a boat (1); It floats (1); Because shape holds air (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERKEY

ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 850 m	2700 - 1850.
2	**(a)** S	The pin falls last.
3	**(d)** metal cup	Metal conducts heat away fastest.
4	**(c)** melting and boiling	States change.
5	**(d)** sand	Insoluble.
6	**(a)** the crystal grows by outside deposition, not from within	Non-living.
7	**(c)** oxygen forms about one-fifth of air	Oxygen is used up.
8	**(a)** waxing	Increasing.
9	**(d)** amount of water	Water amount varies.
10	**(a)** wide field of view to spot predators	Wide view.
11	**(c)** protein and fat	Plant foods have less protein and fat.
12	**(b)** iron only	Only iron is magnetic.
13	**(a)** it is in linear motion and its speed increases	It falls along a straight line, speeding up.
14	**(a)** C conducts heat and D does not	Test for conduction.
15	**(a)** the 20 °C object	Heat enters the colder one.
16	**(c)** 10 mL	50/5.
17	**(c)** 18 g	Half.
18	**(d)** water is needed for germination	Controlled comparison.
19	**(c)** loosening and mixing it	Aeration.
20	**(d)** one sixth	Weaker gravity.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	It represents the Earth in small size (1); It helps us study things too large to observe wholly (1).	1+1
22	It breathes air through lungs (1); It gives birth to young and feeds milk (1).	1+1
23	Lacks carbohydrates and fibre (1); Lacks many vitamins (1).	1+1
24	No (1); Heat weakens the magnet (1).	1+1
25	0.003 km = 3 m (1); 3 m = 300 cm (1).	1+1
26	Rubber is an electrical insulator (1); Prevents electric shock (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Use a flexible tape (1); Follow the curve (1); Verify with thread (1).	1+1+1
28	Prevent shock (1); Rubber or plastic (1); Insulate the hand (1).	1+1+1
29	Reduces loss of hot air (1); Stops steam escaping (1); Keeps heat (1).	1+1+1
30	Moist air rises over hills (1); It cools and condenses (1); Clouds drop rain (1).	1+1+1
31	Immiscible (1); Kerosene lighter (1); Floats on top (1).	1+1+1
	OR: Respiration is very slow (1); It germinates later (1); So it is alive but dormant (1).	1+1+1
32	Half the sheets (1); 100 sheets (1); 100 saved (1).	1+1+1
	OR: Gravity pulls planets (1); Pull bends their path (1); Without it they would fly off in straight lines (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Matka: evaporation (1); Quilt: trapped air (1); Mud house: poor conductor (1); Fridge: removes heat (1); Thermos: insulates (1).	1+1+1+1+1
34	Ice warms to 0 °C (1); Flat at 0 °C: melting (1); Water warms to 100 °C (1); Flat at 100 °C: boiling (1); Heat used for change of state (1).	1+1+1+1+1
35	Sea water collects in pans (1); Evaporation in stages (1); Brine thickens (1); Crystals formed (1); Washed and dried (1).	1+1+1+1+1
36	Roots to water (1); Shoots to light (1); Camouflage (1); Fleeing predators (1); Hibernation or migration (1).	1+1+1+1+1
	OR: Reduce plastic (1); Walk or cycle (1); Save water (1); Recycle (1); Plant trees (1).	1+1+1+1+1
37	Moon's phase depends on position (1); Not on place (1); Rise times differ from place to place (1); Earth rotates (1); The same phase is seen everywhere (1).	1+1+1+1+1
	OR: Sum = 90 s (1); Average = 30 s (1); One swing = 1.5 s (1); Timing many swings reduces error in starting and stopping (1); Dividing gives the time for one swing (1).	1+1+1+1+1
38	Hollow bones: lightness (1); Feathers: flight and warmth (1); Beak: food type (1); Feet: perching, swimming or hunting (1); Examples: duck, eagle, parrot (1).	1+1+1+1+1
	OR: Breakfast: milk, fruit, paratha (1); Lunch: roti, dal, vegetable, curd (1); Snack: sprouts or fruit (1); Dinner: light khichdi with vegetables (1); Avoid chips, soda and excess sweets (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERY